PROJECT UPDATE:

AI WORKLOAD PROFILING ON A RISC-V CORE

FROM GEMM MICROBENCHMARK TO MODEL-AWARE PROFILING

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THE PROJECT: WHAT ARE WE DOING?

A reproducible framework that connects AI workload characteristics to performance on configurable RISC-V hardware

Why profile an AI workload?

- Kernel Bottlenecks
- Performance variability
- The Need for Visibility

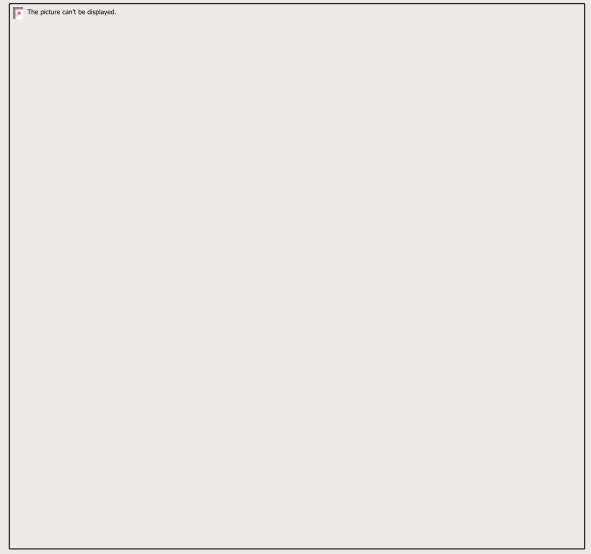
Our profiling pipeline

- Workload
- Simulate
- Analyze
- Optimize

BENCHMARKING WITH GEMM & SIMULATING WITH GEM5

Why GEMM?

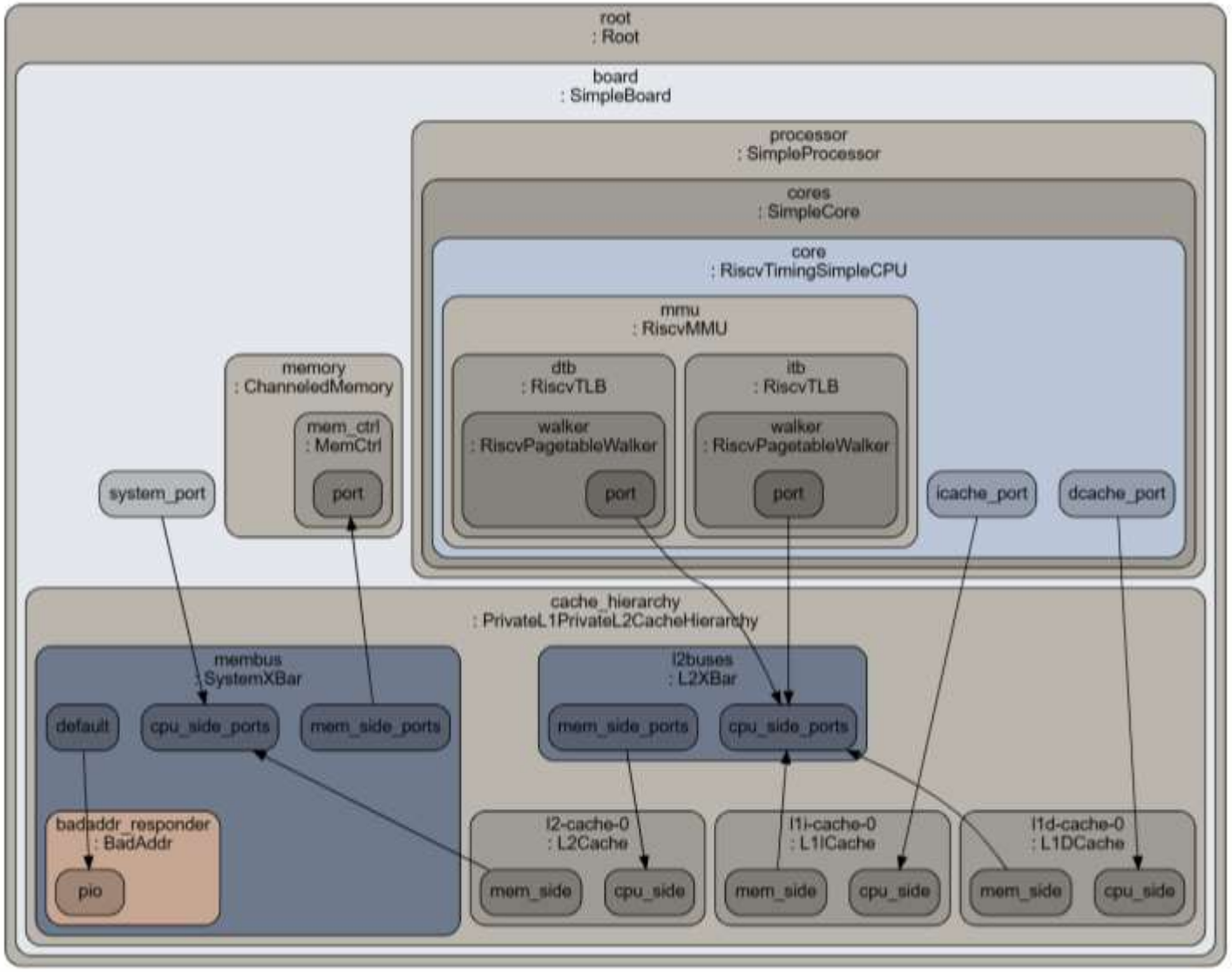
- The core mathematical operation (C = A x B) powering modern AI models, machine learning algorithms, and LLMs.
- Dominates execution time and heavily taxes processor resources.
- Easy to verify, highly optimized, and acts as the ultimate benchmark for processor limits.



Source: nvidia docs

WHY GEM5?

- Simulates physical hardware behavior step-by-step.
- Measures statistics you can't easily see on real boards, like cache misses, execution cycles, and memory stalls.
- Eliminates the need for expensive, hard-to-find physical RISC-V development boards.

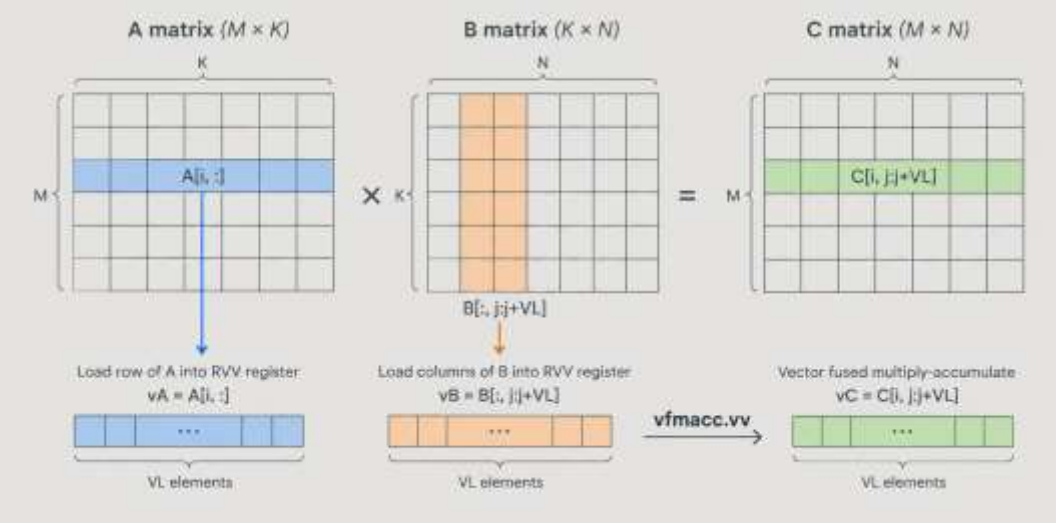


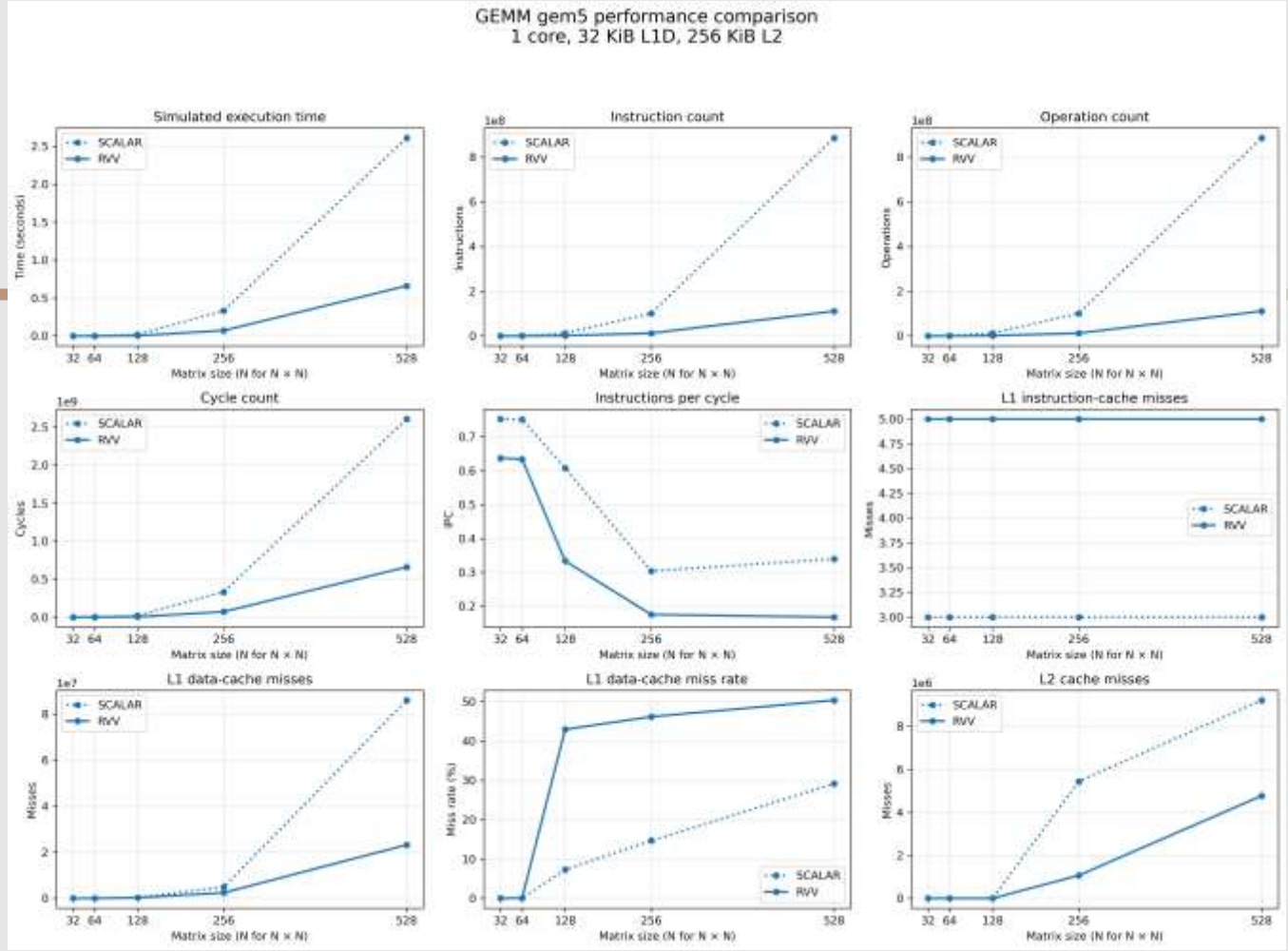
Source: Gem5 documentation

IMPROVEMENTS SINCE LAST UPDATE

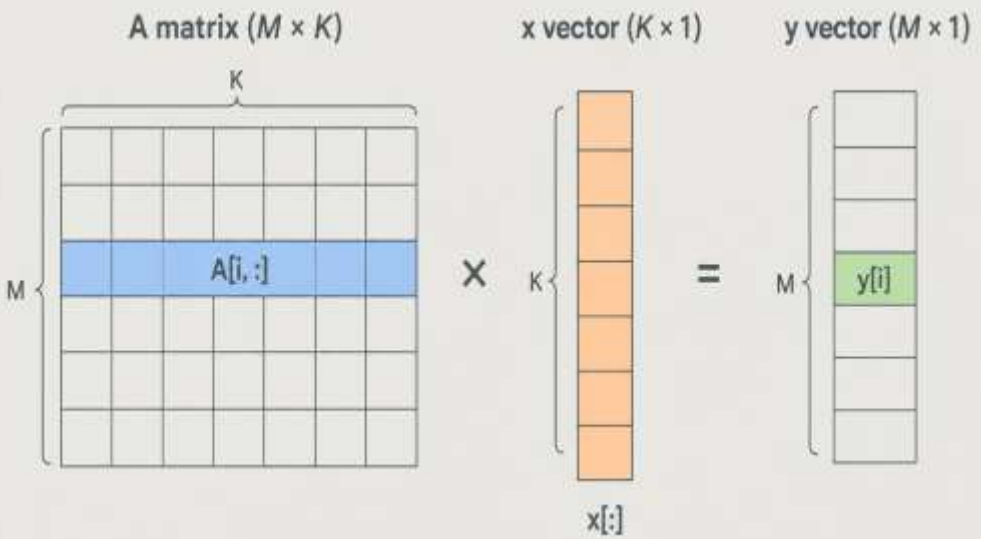
- Vector extension
- Rectangular gemm
- Tiny AI: GPT-2 model

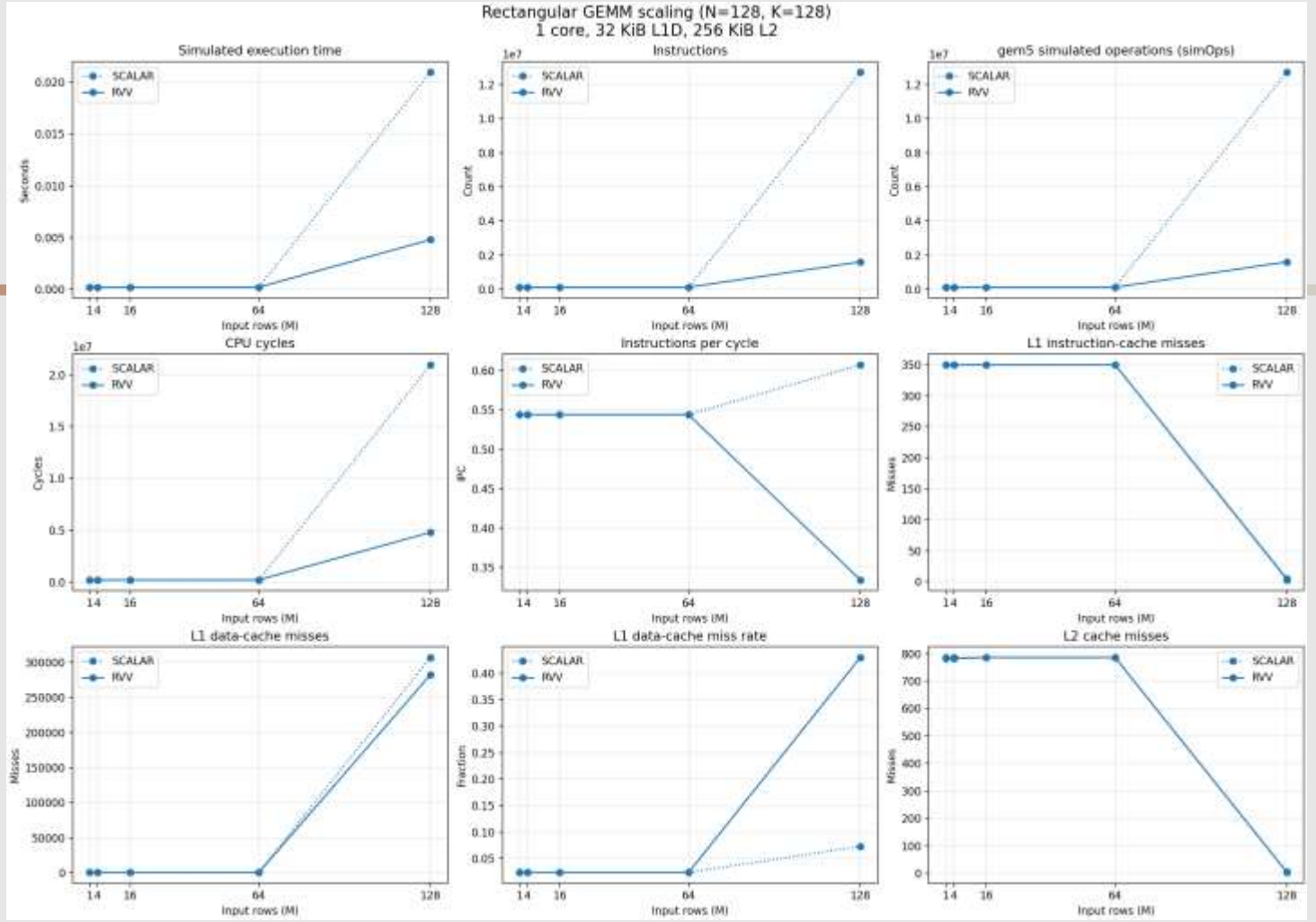
VECTOR EXTENSION



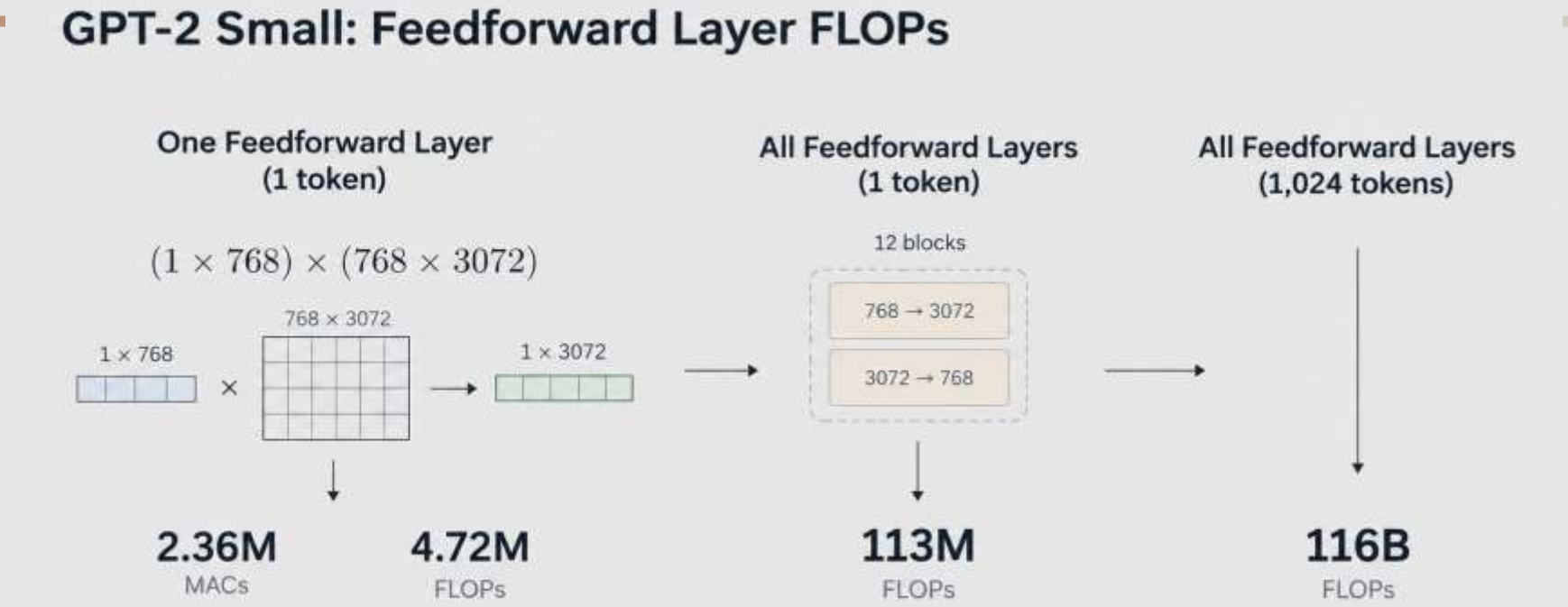


RECTANGULAR GEMM

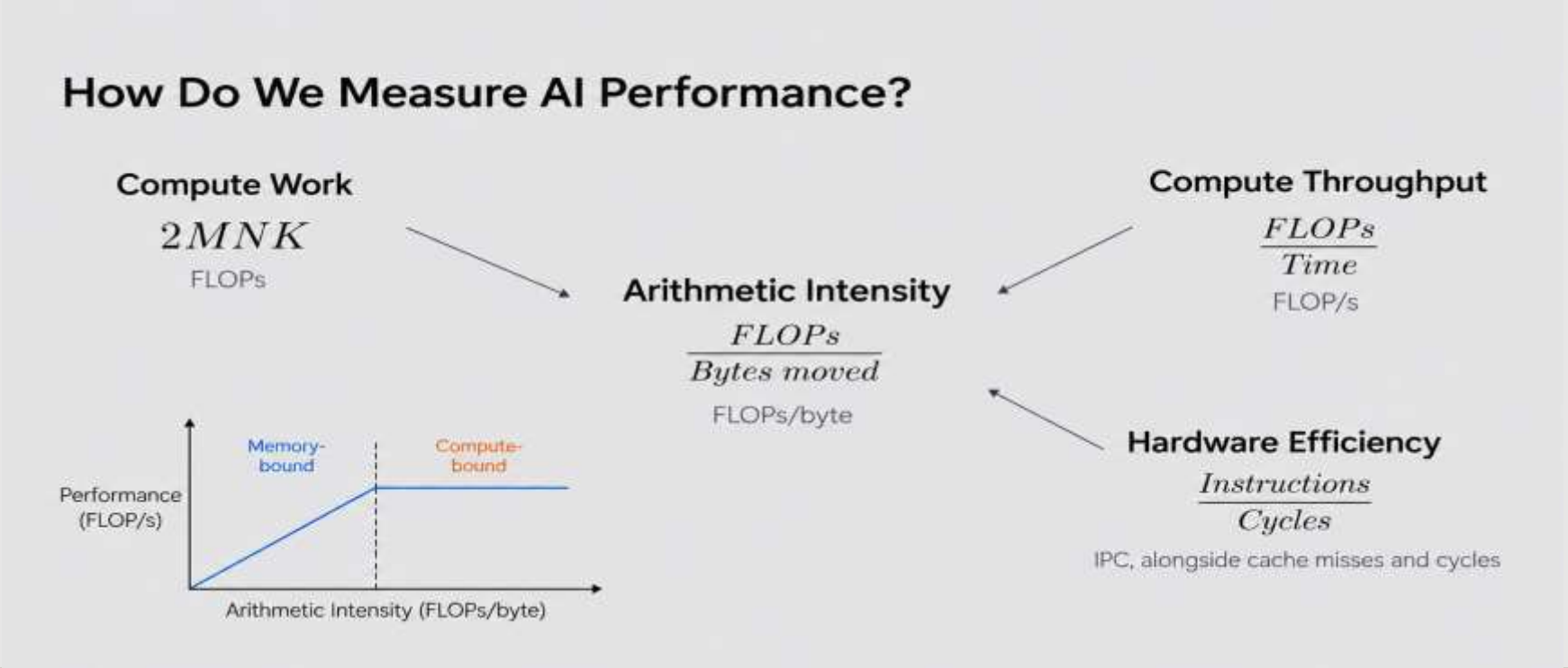




SMALL AI MODEL(GPT-2)



BEYOND GENERIC METRICS: PERFORMANCE EVALUATION



THANK YOU

